SECURITY PRINTING

GRAPHIC DESIGN & MULTIMEDIA PROGRAM

COURSE CODE: GDM5313

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LEVEL: 3

LECTURE 2

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AGENDA

- Document Structure
- Criteria For Selecting Digital Printing Methods
- Card Personalization
- Materials Used To Produce Digital Cards

PERSONALIZATION OF CARDS AND MATERIALS USED TO PRODUCE DIGITAL CARDS

INTRODUCTION

The identification of official documents, whether identity documents, passports, driving licenses, etc., has evolved from handwriting to analog methods such as mechanical copying machines, all the way to the modern digital approach to identification and data printing. This is in addition to the development of the mechanism for adding a personal photo of the card or passport holder. Previously, a photograph was attached using stickers or a metal pin, which facilitated the process of tampering with, changing, or replacing the image.

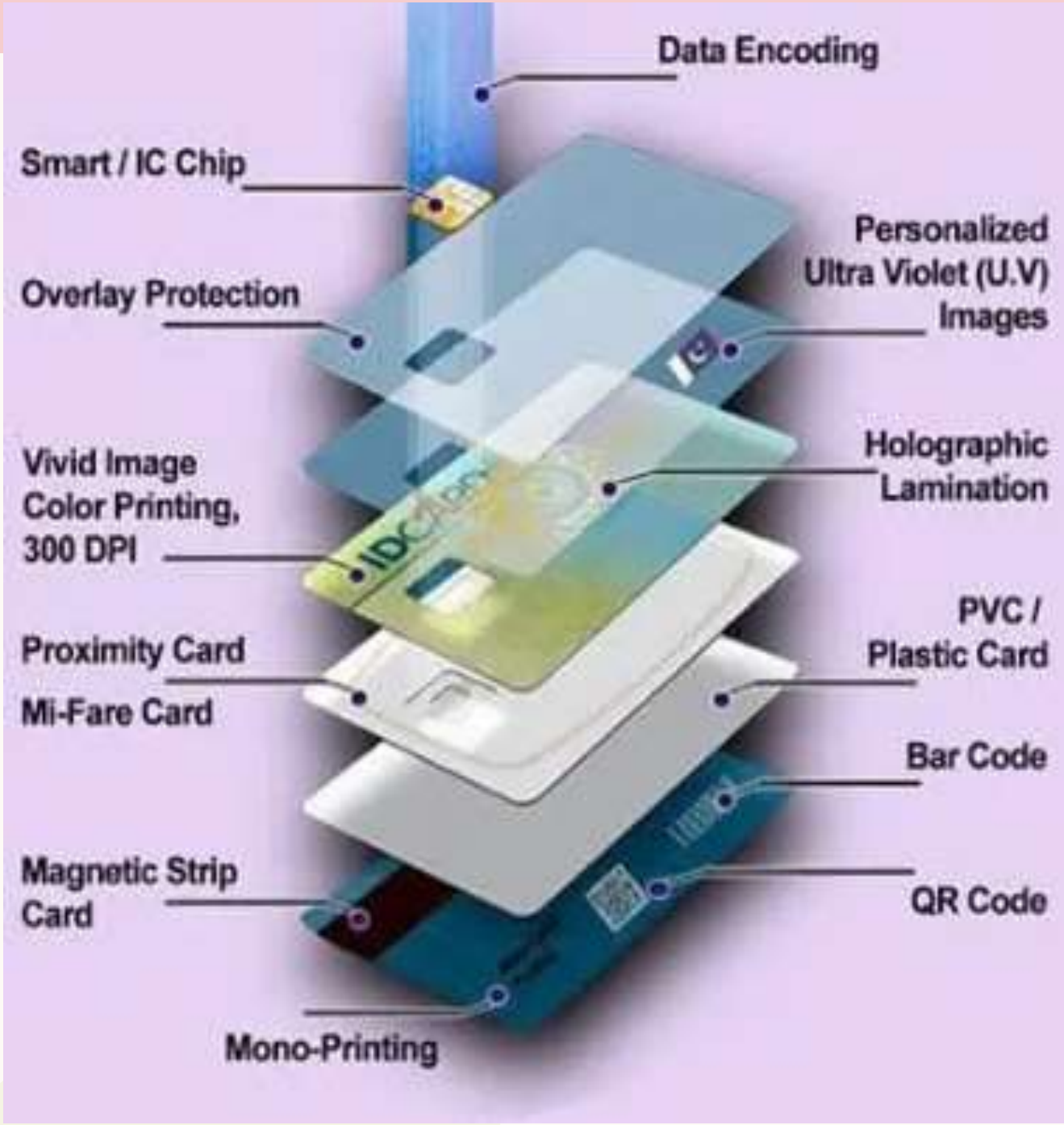
- The method of identifying any document must have two extremely important characteristics that can be considered the cornerstone of the document security and identification process:

- Easy to verify .
- Difficult to reproduce

DOCUMENT STRUCTURE



If we imagine that the structural composition of any secured document consists of several layers stacked on top of each other according to the chronological order of its life cycle, the above figure can give us a simple overview of the composition of the secured document and its various production stages.

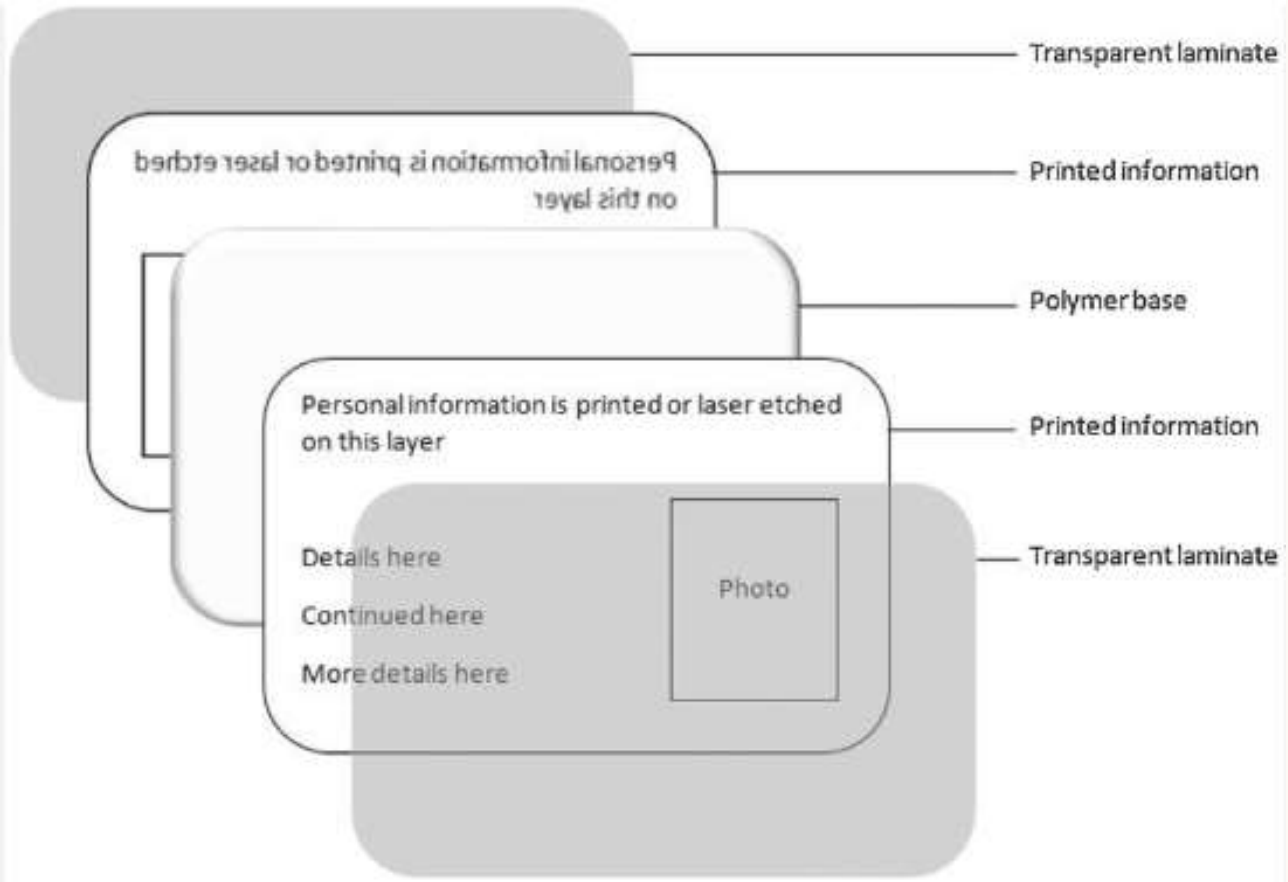


As shown,

the first layer, or the basic foundation of the document's composition, represents the primary material from which the document is made.

This layer can be either natural materials such as paper or synthetic materials such as polymers.

This layer, of course, contains all the essential security elements produced in the manufacturing stages of secured paper or synthetic polymers, such as watermarks, warranty strips, fluorocarbon fibers, and so on.



If we imagine that the structural composition of any secured document consists of several layers that overlap one another according to the chronological order of the document's life cycle, the previous figure can give us a simple idea of the composition of the secured document and its various production stages.

Anatomy of a card

Card Layers



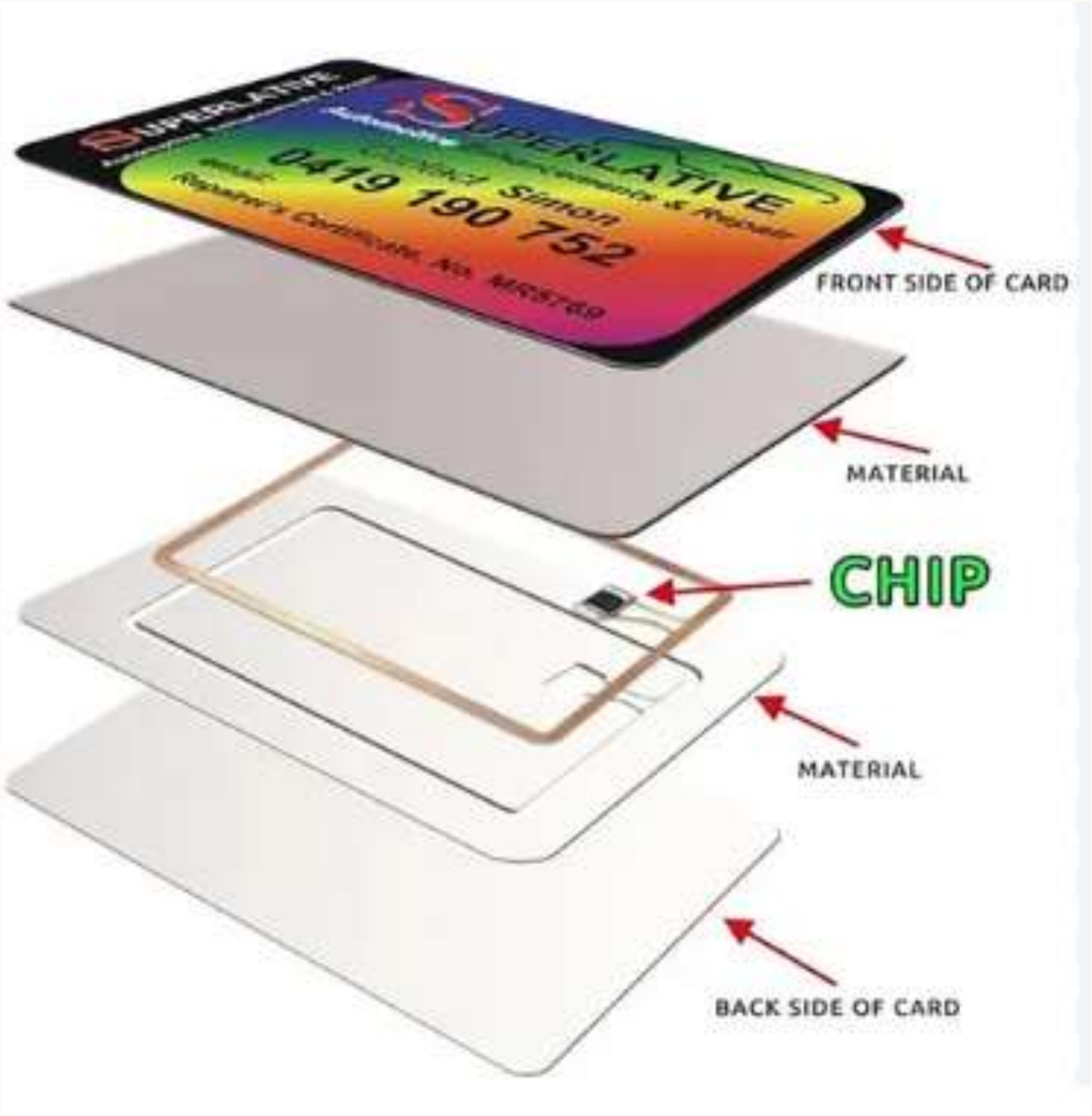
Overlaminate (clear or holographic)

FARGO® HDP Film - cardholder data and variable graphics printed with High Definition Printing™ (HDP®) technology

Clear PVC layer

PET layer containing lithographic and digital printing, and embedded anti-counterfeiting

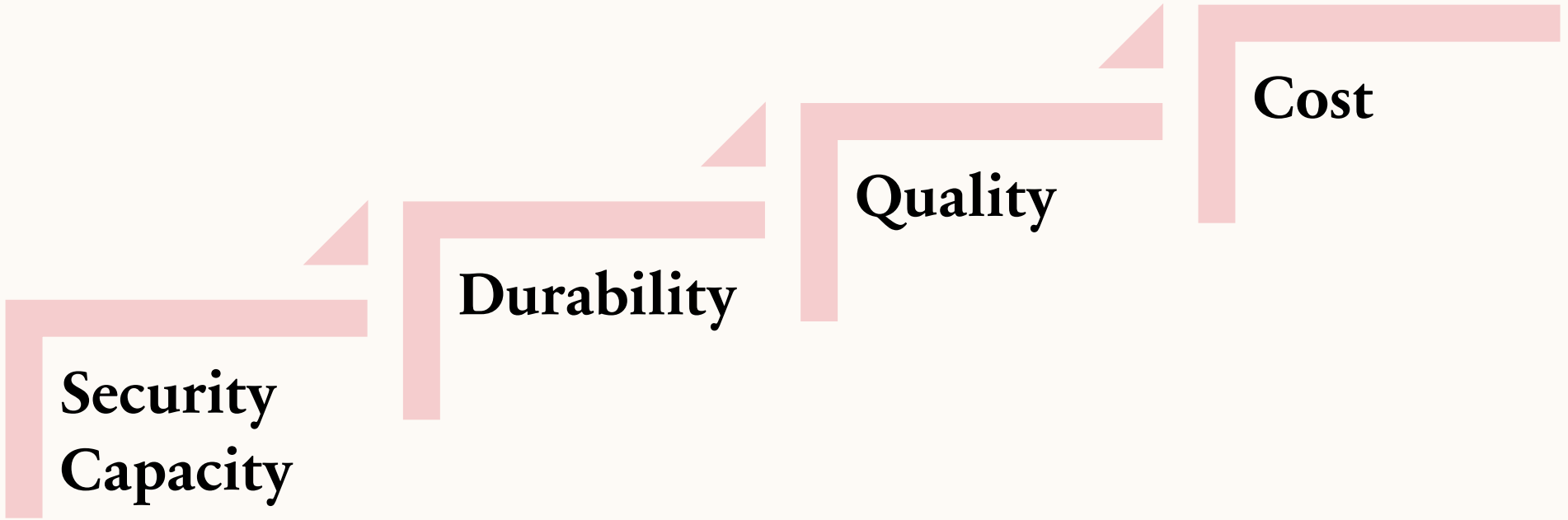
Prelam containing contactless technology (e.g. iCLASS®, Prox, Mifare™, Legic)


FRONT SIDE OF CARD
MATERIAL
CHIP
MATERIAL
BACK SIDE OF CARD

The fourth and final layer may sometimes be unnecessary, and consists of adding transparent panels, or transparent panels with visually variable markings such as transparent holograms. These panels can be either specific in location and dimension, or cover the entire document surface. In this case, they are used to increase security and protect data from tampering, a process somewhat similar to lamination.

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CRITERIA FOR SELECTING DIGITAL PRINTING METHODS USED TO PERSONALIZE PRINTED MATERIALS:



INSURANCE CAPACITY:

When considering a digital printing method for adding an image or variable data, whether on passports, ID cards, or even paper forms, the primary goal should certainly be to consider the insurance value this method can add to the document, rather than diminish its strength.

The method must have insurance features appropriate to the type of document it is adding to, and should not become an element that weakens its insurance value.



DURABILITY

- The digital printing method used for personalization must have appropriate durability, commensurate with the document's service life.
- This is to prevent the printed data or images from being damaged or erased within a period shorter than the document's lifespan.
- All document elements must be integrated and equally durable throughout the document's lifetime.

QUALITY

The personalization of data in secure documents is of utmost importance, as these documents may include identification papers, passports, or other types of documents that grant their holders civil or financial rights. Therefore, the data and images produced must be of the highest quality to ensure maximum verification without ambiguity or confusion.

COST

- Cost is an important factor when deciding on a digital printing method for personalizing or printing a document or document. However, this factor pales in importance next to the security factor. While cost is more important for short-life or single-use prints, such as railway tickets, venue tickets, or match tickets, the cost factor for securing important documents, such as credit cards, passports, or identity cards, takes a backseat to security, durability, and quality, given the overall value of the document.
- This value can be related to national security, such as ID cards or passports, or it can be financial, such as credit cards.

CARD PERSONALIZATION

Digital cards are not just what can be described as ID cards,

but rather anything in the form of a card or card with specific dimensions specified in international quality standards through the ISO standard.

This standard concerns the physical description of card dimensions as follows:

السمك (الثخانه)		الارتفاع		العرض		نوع البطاقة
Min.	Max.	Min.	Max.	Min.	Max.	
0.68	0.84	14.90	15.10	24.90	25.10	ID-0
0.68	0.84	53.92	54.18	85.47	85.90	ID-1
0.68	0.84	74.70	74.30	104.80	105.30	ID-2
0.68	0.84	87.70	88.30	124.80	125.30	ID-3

MATERIALS USED TO PRODUCE DIGITAL CARDS

The materials used to produce digital cards have evolved into their modern form, beginning with secure paper.

Beginning in the 1970s, plastic materials assumed a significant position in the production of secure documents, due to their greater security, reliability, and durability compared to paper materials. During this period, a wealth of accumulated experience and lessons learned enabled plastic materials to occupy the leading position in the production of digital cards.

Polycarbonate also accepts printing in multiple ways, and optically variable laminates, such as holograms or kinogram, can be added to it, helping to secure cards comprehensively. One of the major drawbacks of polycarbonate is its tendency to become brittle with frequent use, and its surface is susceptible to cracking and damage, which can lead to card body breakage. However, this doesn't mean that polycarbonate is the only material used to produce cards; many other plastic materials are used to produce various types of cards.


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0:00 / 1:59 • Introduction

The most important of these materials are:

- Polyvinyl chloride
- Polyester
- Composite materials

POLYVINYL CHLORIDE (PVC)

This is the least expensive material in the family of plastic materials known for producing digital cards. It is widely used in the manufacture of communication cards and contactless cards. In general, the operational life of these cards is relatively shorter than that of ID cards or vehicle and driver's licenses due to the lower resistance of these materials to heat, loads, bending, and ultraviolet rays, which can cause premature deterioration of these types of cards through layer separation, partial damage, or surface erosion. The expected lifespan of these types of cards is between 3-5 years.

POLYESTER

- The properties of polyester have been improved, and some modifications have been made to this material to achieve higher quality specifications for digital card production.
- The most important of these modified materials PETG and PETF This aims to reduce weaknesses and improve polyester properties, as well as to improve the physical durability of the card body and its resistance to bending and warping. The new types also withstand high temperatures and have excellent solvent resistance.

COMPOSITE MATERIALS

Card types are produced using more than one material in a single configuration to achieve distinct properties that cannot be achieved from a single material, thus achieving competitive properties in terms of quality and affordability.

ASSIGNMENT TASK

IN A TABLE COMPARE BETWEEN THE EGYPTIAN ID CARD AND ANY OTHER ID CARD IN THE SECURITY FEATURES, CHARACTERISTICS AND DESIGN